Investigating the Effect of Innovation Activities of Firms on Innovation Performance: Does Firm Size Matter?

by

Mahour Parast
Arizona State University

Adegoke Oke
Arizona State University

Matthew Doolin
Duke University

CES 25-04 **January 2025**

The research program of the Center for Economic Studies (CES) produces a wide range of economic analyses to improve the statistical programs of the U.S. Census Bureau. Many of these analyses take the form of CES research papers. The papers have not undergone the review accorded Census Bureau publications and no endorsement should be inferred. Any opinions and conclusions expressed herein are those of the author(s) and do not represent the views of the U.S. Census Bureau. All results have been reviewed to ensure that no confidential information is disclosed. Republication in whole or part must be cleared with the authors.

Papers in the series are posted on www.census.gov/library/working-papers/series/ces-wp.html. For information about the series, contact Sean Wang, Editor, Discussion Papers, U.S. Census Bureau, Center for Economic Studies, 4600 Silver Hill Road, Washington, DC 20233, CES.Working.Papers@census.gov.

Abstract

Understanding the relationship between a firm's innovation activities and its performance has been of great interest to management scholars. While the literature on innovation activities is vast, there is a dearth of studies investigating the effect of key innovation activities of the firm on innovation outcomes in a single study, and whether their effects are dependent on the nature of firms, specifically firm size. Drawing from a longitudinal dataset from the Business Research & Development and Innovation Survey (BRDIS), and informed by contingency theory and resource orchestration theory, we examine the relationship between a firm's innovation activities - including its Research & Development (R&D) investment, securing patents, collaborative R&D, R&D toward new business areas, and grants for R&D - and its product innovation and process innovation. We also investigate whether these relationships are contingent on firm size. Consistent with contingency theory, we find a significant difference between large firms and small firms regarding how they enhance product innovation and process innovation. Large firms can improve product innovation by securing patents through applications and issuances, coupled with active participation in collaborative R&D efforts. Conversely, smaller firms concentrate their efforts on the number of patents applied for, directing R&D efforts toward new business areas, and often leveraging grants for R&D efforts. To achieve process innovation, a similar dichotomy emerges. Larger firms demonstrate a commitment to securing patents, engage in R&D efforts tailored to new business areas, and actively collaborate with external entities on R&D efforts. In contrast, smaller firms primarily focus on securing patents and channel their R&D efforts toward new business pursuits. This nuanced exploration highlights the varied strategies employed by large and small firms in navigating the intricate landscape of both product and process innovation. The results shed light on specific innovation activities as antecedents of innovation outcomes and demonstrate how the effectiveness of such assets is contingent upon firm size.

Keyword: R&D investment, firm size, product innovation, process innovation, patent, collaborative R&D, contingency theory, resource orchestration theory

---

[*] Disclaimer. Any views expressed are those of the authors and not those of the U.S. Census Bureau. The Census Bureau's Disclosure Review Board and Disclosure Avoidance Officers have reviewed this information product for unauthorized disclosure of confidential information and have approved the disclosure avoidance practices applied to this release. This research was performed at a Federal Statistical Research Data Center under FSRDC Project Number 1577. (CBDRBFY22-253).

Acknowledgment. This research is supported by a grant from the National Science Foundation (NSF).
Conflict of Interest Statement. The authors do not have any conflicts of interest to disclose.

Parast: Ira A. Fulton Schools of Engineering, Arizona State University, Tempe, AZ. Oke: W. P. Carey School of Business, Arizona State University, Tempe, AZ. Doolin: Social Science Research Institute, Duke University, Durham, NC.

1. Introduction

Innovation is widely regarded as the main driver of economic growth and prosperity. According to scholars in management and economics, firms can achieve a competitive advantage through emphasizing innovation (e.g., Schumpeter, 1934; Singh et al., 2017; Jajja et al., 2017; Carnes et al., 2022). Management scholars have long acknowledged the centrality of innovation and change as one of the most vital and demanding challenges in organizations (Tushman and Nadler, 1986; Damanpour, 2001; Wells, 2008; Bauer and Leker, 2013; Sabahi and Parast, 2020). As such, several management programs such as Business Process Reengineering (BPR), Total Quality Management (TQM), Lean Production, Simultaneous Engineering, and Just-in-Time Production (JIT) have been embraced by a wide variety of businesses to accelerate the development of new products and services as well as improvement in organizational processes (Kim et al., 2012; Ding et al., 2023; Frank et al., 2024). While there is clear convergence among researchers and practitioners that innovation is a source of competitive advantage (Rosenbusch et al., 2011; Gao et al., 2023), the specific antecedents of successful new products, services, and processes remain a challenge for organizations (Griffin, 1997; Pearce and Ensley, 2004; Parast, 2011; Cesaroni and Piccaluga, 2013; Prajogo and Oke, 2016).

In this study, our focus is to examine the effect of a firm's *innovation activities* on innovation outcomes and to determine whether firm size matters. Prior studies have primarily examined the effect of R&D investment and the number of patents on firm performance (Artz et al., 2010; Andries and Faems, 2013). However, little is known about how innovation activities, specifically R&D efforts toward new business areas, collaborative R&D, and grants for R&D, impact firm innovation outcomes such as product innovation and process innovation. We take a much broader approach to understanding the effect of a firm's innovation activities on the

development of new products, services, and processes, which encompass both internal resources and external engagement with partner organizations. In addition, whether firm size has an impact on the effectiveness of innovation activities of the firm remains relatively unexplored. We aim to address these gaps in the literature by examining a broad set of innovation activities of the firm, including R&D investment, securing patents, R&D toward new business, collaborative R&D, and grants for R&D, to provide a more nuanced assessment of the relationship between R&D activities and innovation outcomes.

Innovation is a multi-dimensional construct (Mabenge et al., 2022), comprising service innovation, technological innovation, business model innovation, and the like (Geissdoerfer et al., 2018; Cheng and Sheu, 2023). In this study, we focus on product innovation and process innovation, which have been regarded as two distinct and complementary innovation outcomes that enable firms to achieve a competitive advantage (Voss, 1994; Choo et al., 2021; Kesidou et al., 2022; Mariani et al., 2023). Thus, the first objective of the study is to investigate the effect of innovation activities (R&D investment, securing patents, R&D toward new business areas, collaborative R&D, and grants for R&D) on product innovation and process innovation. We draw from the resource orchestration theory (Sirmon et al., 2007, 2011) to argue that R&D investment does not lead to strengthening innovation outcomes by itself; rather, it is the implementation of key innovation or knowledge management practices that serves as the locus of innovation.

To gain a full understanding of the nuanced relationships between innovation activities and innovation performance, we investigate whether these relationships are contingent upon firm size. There is a substantial body of literature indicating that small and large firms exhibit differences in their innovative behaviors and resulting outcomes (Cohen and Klepper, 1996a; Cohen and Klepper, 1996b; Shefer and Frenkel, 2005; Berends et al., 2014). We aim to provide a more

nuanced assessment of such relationships. Building upon the contingency theory of management, the second objective of this study is to investigate whether firm size has a significant impact on the relationships between innovation activities and innovation performance (Damanpour, 1991; Camison-Zornoza et al., 2004; Revilla and Fernández, 2012). Specifically, we examine whether the relationships between innovation activities (R&D investment, securing patents, R&D toward new business areas, collaborative R&D, and grants for R&D) differ in small and large firms. Understanding the significant drivers of innovation outcomes such as product innovation and process innovation is critical for organizations as it sheds light on which aspects of their innovation activities are consequential.

2. Theoretical Perspectives: Contingency Theory and Resource Orchestration Theory

The relationship between a firm's innovation activities and innovation outcomes is examined from two perspectives: contingency theory and resource orchestration theory. We discuss these two theoretical perspectives and articulate how they help us understand the relationship between innovation practices and innovation outcomes.

The contingency theory of management addresses the effect of organizational, contextual, and environmental factors on a firm's performance (Woodward, 1958; Burns and Stalker, 1961; Chandler and Chandler, 1962). To respond to a myriad of organizational and contextual factors and to align themselves with the evolving nature of their business environment, organizations may need to emphasize different sets of practices to better align with their environment (Donaldson 2001).

Although contingency theory has been predominantly used to explain how environmental factors influence relationships between phenomena, its role in explaining the effect of organizational factors such as firm size has been recognized. LiPuma et al. (2013) reported that

institutional quality was more important in increasing the export performance of new and small firms compared with large, established firms, suggesting that the relationship between institutional quality and export performance is contingent upon firm size. Empirical studies show that small and medium-sized firms seem quite similar in their innovation management practices, whereas large firms exhibit more distinct behavior (Vaona and Pianta, 2008), suggesting that size may play an important role in understanding the differences in innovation management practices of firms.

Resource orchestration theory (ROT) provides a useful and complementary perspective to explain the development of new products and processes by organizations. ROT asserts that management practices, through structuring, bundling, and leveraging organizational resources, can create a sustainable competitive advantage (Sirmon et al., 2011). Building upon the resource-based view (RBV), ROT identifies processes by which an organization can integrate valuable resources across the organization to improve performance. RBV is concerned with organizational resources that are rare, valuable, and inimitable; ROT is primarily concerned with the process by which an organization can integrate and mobilize these resources to achieve superior performance (Barney, 1991; Sirmon et al., 2007; Sirmon et al., 2011). Such processes could include practices required for the coordination of organizational resources. ROT is particularly relevant in explaining the relationship between innovation investment and activities and innovation performance because it is how the assets are structured and leveraged that make them impact innovation performance.

3. Hypotheses: Firm Size, Innovation Investment and Activities, and Innovation Outcomes

Scholars and practitioners have shown great interest in identifying organizational, contextual, and environmental factors that impact organizational outcomes such as product innovation and process innovation (Cohen and Klepper, 1996a; Cohen and Klepper, 1996b; Visnjic et al., 2016; Xie et al., 2019; Chirumalla, 2021). However, the results of multiple empirical

studies on contextual factors that impact innovations in organizations have been mixed (Downs and Mohr, 1976; Kimberly and Evanisko, 1981; Crossan and Apaydin, 2010). The contingency view of innovation helps us to examine the contingency and contextual factors that impact the relationship of knowledge assets such as R&D investment and innovation outcomes (Damanpour and Aravind, 2006; Ahmed et al., 2020). We identify firm size as an important contingency factor and examine its influence on the relationship between innovation activities and innovation outcomes. More specifically, we argue that the effects of innovation activities (i.e., R&D investment toward new businesses, collaborative R&D, securing patents, and grants for R&D) on innovation outcomes vary between small firms and large firms.

3.1. Firm Size and Firm Innovation Outcomes

Firm size is defined in terms of an organization's resources, turnover, or workforce size. Lee and Xia (2006, p. 975) contend that firm size can be seen as a surrogate for total resources and slack resources.

Firm size has been empirically linked to innovation. For example, there is a positive relationship between firm size and R&D expenditure, suggesting that larger firms tend to invest more in R&D (Shefer and Frenkel, 2005). Such investments stimulate innovation; thus, larger firms are expected to be more innovative in creating new knowledge (e.g., patents) or innovation outcomes (e.g., product innovation or process innovation). However, with size comes complexity; larger firms tend to have more complex networks or supply chains, which limit the ability to manage organizational processes and routines. Complex networks also limit access to a wide range of resources within the organization, across the supply chain, or in the strategic alliance network (Kimberly, 1976; Josefy et al., 2015). This added complexity could undermine the effectiveness of the innovation process. Indeed, Shaver and Mezias (2009) argue that as firm size grows, there

is a non-proportional increase in the resources required for the efficient management and coordination of complex operations. As the organization grows in size, internal coordination becomes more intricate and more costly to manage (Wickert et al., 2016). While larger firms benefit from economies of scale and access to additional resources for innovation, the coordinating costs could constitute an impediment. For larger firms, complex organizational processes and rigid structures may limit the efficiency and effectiveness of new product launches. Alternatively, smaller organizations have the disadvantage of limited access to resources; however, due to less complex organizational processes and routines, they can be more efficient and effective in innovating (De Jong and Marsili, 2006; Leal-Rodríguez et al., 2015).

The relationship between firm size and innovation has been extensively discussed in management literature (Damanpour, 1992; Knot and Vieregger, 2020). Studies provide mixed findings on the effect of firm size on different innovation outcomes. While it is generally believed that large firms tend to emphasize process innovation more than product innovation, and smaller firms benefit more from product innovation, it is unclear which innovation practices or activities of a firm can lead to product innovation or process innovation. Larger firms have a significant advantage over smaller firms in terms of their higher level of technical efficiency, allowing them to enjoy more marginal cost improvements through investment in process innovation initiatives (Lundvall and Battese, 2000; Alvarez and Crespi, 2003; Soderbom and Teal, 2004). Additionally, Cohen and Klepper (1996a) demonstrate that as the size of a firm increases, the potential payoff from engaging in process innovation increases; therefore, larger firms benefit more from process innovation activities. Cost reduction is one of the key benefits of process innovation. Large firms are more likely to seek cost reduction due to their relatively high operating costs compared to small firms, suggesting that large firms are likely to emphasize process innovation more. Conversely,

smaller firms have been shown to excel more in product and business innovation (Wagner and Hansen, 2005). The path to faster growth is more likely through product innovation, which directly contributes to an increase in revenue compared to process innovation (Braguinsky et al., 2021). Smaller firms tend to be more aggressive in seeking growth and are, therefore, more likely to emphasize product innovation than process innovation. Taken together, we offer the following hypotheses to depict the relationships between size and different innovation outcomes in firms.

H_{1a}: Smaller firm size is more positively related to product innovation.

H_{1b}: Larger firm size is more positively related to process innovation.

3.2. Patents and Firm Innovation Outcomes

Studies have discussed the difference between small and large firms in their patenting activities. Small firms can benefit from patenting through licensing their innovations and generating revenue (Andries and Faems, 2013). Pavitt (1985) found that small firms tend to secure more patents per unit of R&D expenditure compared to their larger counterparts. This observation has given rise to an argument proposed by some scholars, such as Plehn-Dujowich (2009), that small firms generally demonstrate greater patent levels.

We make a distinction between two dimensions of patents: *patents applied for*, and *patents issued.* While most studies in the literature on innovation management use the number of patents issued, we examine both dimensions of the firm's patenting strategies for two reasons. First, from a legal standpoint, there is a difference between these two dimensions of patenting. While a firm can apply for many patents, it cannot gain protection for an innovation until the patent is issued. Second, a small business or a start-up firm may be inclined to apply for patents to demonstrate its capability for knowledge generation, which can be appealing to investors and can accelerate small business growth.

It should be noted, however, that a patent application or a patent issued does not necessarily lead to an innovation outcome (i.e., a marketable product or a new process that provides a competitive advantage). According to the resource orchestration theory, firm resources are integrated and bundled to convert patents into innovation outcomes. The decision to patent or retain the patent depends primarily on the business objectives. Drawing from contingency theory, the patenting strategy of firms may depend on the resources available to them, which is a function of the size of the organization. Athreye et al. (2019) discuss the differences between small firms and large firms in terms of the value of patents; large firms tend to produce a higher number of innovations compared to small firms, increasing the likelihood of such innovations emanating from patents.

Previously, we argued that smaller firms are more likely to emphasize product innovation while larger firms tend to prioritize process innovation. As such, it could be argued that smaller and larger firms are likely to focus more on patents for product and process innovation respectively. One of the key motivations for small firms' patenting strategy is licensing to attract investment; this is easier to do with product innovations, which are more easily visible compared to process innovations that are typically hidden from the marketplace. As such, we anticipate that small firms will predominantly pursue patents related to product innovation, whereas large firms will focus on patents targeting process innovation leading to the following hypotheses.

> *H_{2a}: There is a significant difference in the effect of the number of patents applied for on product innovation between small and large firms. The effect is expected to be more pronounced for small firms.*

H_{2b} There is a significant difference in the effect of the number of patents applied for on process innovation between small and large firms. The effect is expected to be more pronounced for large firms.

H_{2c}: There is a significant difference in the effect of the number of patents issued on product innovation between small and large firms. The effect is expected to be more pronounced for small firms.

H_{2d}: There is a significant difference in the effect of the number of patents issued on process innovation between small and large firms. The effect is expected to be more pronounced for large firms.

3.3. Collaborative R&D

Research shows that collaborative R&D provides many advantages, such as pooling complementary capabilities and facilitating learning between partners (Vlaisavljevic et al., 2022). However, collaborative R&D requires coordination, monitoring, and knowledge-sharing procedures, as well as the need to address potential opportunistic behavior by partners (Hottenrott and Lopes-Bento, 2016). As such, significant resources, which may not be immediately available to small firms, are required for collaborative R&D; this suggests that larger firms benefit more from collaborative R&D.

The effects of collaborative R&D are not uniform (Freel, 2003) and may be contingent on some factors as argued by the contingency theory. Recognizing that small and large firms tend to follow distinct innovation processes, it is reasonable to expect that the impact of technological collaboration will differ based on firm size and the specific innovation outcomes the collaboration aims to achieve. Engaging in collaborative R&D efforts requires the trade-off between managing coordination and oversight, the demands of the disclosure of knowledge, and it entails the potential

risk of opportunistic behavior by the partnering entities (Hottenrott and Lopes-Bento, 2016). Such risks are expected to be much larger for small firms, which may weaken the capability of small firms to benefit from collaborative R&D efforts to generate innovation outcomes. Thus, we propose these hypotheses:

H_{3a}: There is a significant difference in the effect of collaborative R&D on product innovation between small and large firms. The effect is more pronounced for large firms.

H_{3b} There is a significant difference in the effect of collaborative R&D on process innovation between small and large firms. The effect is more pronounced for large firms.

3.4. R&D Toward New Business

One of the R&D strategies firms can pursue is to explore new business opportunities (Hoang and Rothaermel, 2010; Zhong et al., 2022). The literature provides insights into how small firms and large firms benefit differently from such initiatives (McNamara and Baden-Fuller, 2007). R&D toward a new business (or new businesses) is the result of a firm pursuing a more diversified business strategy. Due to resource availability, larger firms are more likely to engage in and benefit more from diversification of R&D (Aw and Batra, 1998). According to the resource orchestration theory, large firms possess the capability to exploit and orchestrate resources in optimal ways to achieve innovative outcomes compared to small firms that have resource constraints. Large firms are also able to absorb potential risks involved in R&D investment in new businesses compared to small firms.

Research and development activities require immediate expenses, yet returns from such investments take time to materialize. Compared to large firms, small firms have restricted working capital and cannot afford the luxury of waiting for several years for their R&D efforts to generate profits. Consequently, many of a small firm's R&D endeavors are geared towards short-term goals

rather than new businesses. In addition, the uncertainty surrounding the success of diversified R&D endeavors might restrict investment in new businesses by small firms. With their limited available capital, small firms cannot afford to finance numerous high-risk projects (Hogan and Chirichiello, 1974; Marcelino-Sádaba et al., 2014).

For large firms, a diversification strategy serves as a catalyst for accumulating strategic slack. This strategic slack represents a valuable reservoir of knowledge that can be effectively harnessed to stimulate innovation (Orlando et al., 2018). In fact, the effectiveness of R&D investments has been shown to be more favorable for large organizations. Schumpeter (1950) indicated that larger firms, at first, appear to have a higher propensity for innovation, particularly in informal R&D activities. This tendency is attributed to their access to a larger pool of human resources, economies of scale, substantial market influence, financial capacity, and capabilities for profiting from their innovations. This assertion has been supported by recent research studies (Ghosh, 2009; Sasidharan and Kathuria, 2011; Baumann and Kritikos, 2016). Given that R&D activities are inherently risky decisions, pursuing an R&D investment decision in a new business (or technology) domain would be less risky for large firms but riskier for small businesses. Thus, we propose these hypotheses:

H_{4a}: There is a significant difference in the effect of R&D toward new business on product innovation between small and large firms. The effect is more pronounced for large firms.

H_{4b} There is a significant difference in the effect of R&D toward new business on process innovation between small and large firms. The effect is more pronounced for large firms.

4.0 Methodology

We used a longitudinal dataset from the Business R&D and Innovation Survey (BRDIS) to examine the hypotheses. As part of a joint effort between the National Science Foundation

(NSF) and the U.S. Census Bureau, BRDIS collects data on firm innovation practices. To ensure the confidentiality of the respondents (organizations), BRDIS microdata constitute federally protected data that can only be accessed by researchers at one of the U.S. Federal Statistical Research Data Centers (FSRDC), which are managed by the U.S. Census Bureau. BRDIS was sent to a nationally representative sample of all companies with five or more employees in all industries, regardless of whether the company performed or funded R&D. It is estimated that the BRDIS survey was mailed to approximately 70,000 firms. The response rate of BRDIS is reported to be about 77 percent (Knott and Vieregger, 2020).

4.1 Variables and Measures

We use measures of a firm's R&D and innovation activities to examine how the effect of innovation activities on innovation outcomes differs between small firms and large firms. Given that R&D activities and patents capture different perspectives of the innovation process (Licht and Zoz, 2000), such an approach provides a more holistic understanding of the determinants of product innovation and process innovation; it also builds on the finding that firms' innovation activities lead to product innovation and process innovation (Cozzarin, 2017). We distinguish between R&D expenditure and patents: R&D expenditure is a resource utilized for the innovation process, while patents are the outcome of effective utilization of such resources to create new knowledge. It is the combination of these two complementary aspects of innovation activities that leads to product innovation and process innovation.

4.2. Control Variables

To ensure that our findings are consistent across different industry segments, we used industry segments as a control variable. We defined 12 industry segments based on their NAICS coding and the nature of their operations. In addition to industry segments, we have two other sets

of control variables. Because of the longitudinal (panel) design, each firm serves as its own control, allowing us to control for unobserved effects. Finally, we accounted for the different years of observation (2008-2016) by defining nine variables to capture the year effect.

4.3. Dependent Variables

We have two dependent variables. They capture two different types of innovation outcomes (Table 1).[1]

Table 1. Definitions of Product Innovation and Process Innovation in BRDIS

Dependent Variable(s)	Item	Data
Product Innovation	Introducing new or significantly improved goods	Binary
	Introducing new or significantly improved services	Binary
Process Innovation	Introducing new or significantly improved methods of manufacturing or producing goods or services	Binary
	Introducing new or significantly improved logistics, delivery or distribution methods for your inputs, goods, or services	Binary
	Introducing new or significantly improved supporting activities for your processes, such as maintenance systems or operations for purchasing, accounting, or computing	Binary

Product innovation. This is defined and measured in two ways: 1) Introducing new or significantly improved goods, and 2) Introducing new or significantly improved services. The innovation could be new to the firm and/or new to the market. The answers to these questions are 'Yes' or 'No'. Our measure of product innovation is a binary value of one (if the firm reported an innovation) or zero (if there was no innovation).

Process innovation. This is defined and measured in three ways: 1) introducing new or significantly improved methods of manufacturing or producing goods or services, 2) introducing new or significantly improved logistics, delivery, or distribution methods for your inputs, goods, or services, and 3) introducing new or significantly improved supporting activities such as

[1] These variables are extracted directly from the Business R&D and Innovation Survey (BRDIS).

maintenance systems or operations for purchasing, accounting, or computing. The answers to these questions are 'Yes' or 'No'. Our measure of process innovation is a binary value of one (if the firm reported an innovation) or zero (if there was no innovation).

4.4. Independent Variables

R&D Investment. Questions relevant to R&D investment have been incorporated into the BRDIS. The amount of worldwide expenses put toward R&D ($) by firms is used as a measure of R&D investment. Therefore, our unit of analysis is at the firm level.

Firm size. The number of employees is used as a measure of firm size. We divided firms into two categories: small and large. Firms with fewer than 500 employees are considered small, while firms with 500 employees or more are considered large. This classification is based on criteria from the U.S. Small Business Administration (SBA), which defines small and large firms using this cutoff point.

Securing patent. This is an innovation activity that comprises patents applied for and patents issued, which are measured as follows.

Patents applied for: To measure this variable we used the natural logarithm of the number of patents a firm applied for each year. Patents applied for capture knowledge creation capabilities by organizations.

Patents issued. To measure this variables we used the natural logarithm of the number of patents issued for the firm each year. There is an important distinction between patents applied for vs. patents issued; the distinction is mostly related to the effectiveness of the innovation management systems in creating new and valuable knowledge that can be protected and could have the potential to be commercialized.

R&D toward new business areas. This measures the percentage of R&D that is allocated for new business areas. This measure provides an indication of how much a firm is emphasizing exploration in R&D activities.

Collaborative R&D. This measures whether a firm was involved in collaborative R&D activities. This is reported as a binary variable (Yes or No).

Grants for R&D. This measures the R&D paid for by government or private foundation grants. This is reported as a binary variable (Yes or No).

4.5 Data

Observations from 2008-2016 BRDIS data comprise the analysis dataset. Although post-2016 data was available, the survey after 2016 removed the vital questions for this research regarding product innovation and process innovation. The primary focus of the data cleanup was to include as many observations as possible and retain a balanced panel. To avoid dropping observations with missing data values, estimates produced by the Census Bureau were used to replace non-reported values of product innovation and process innovation, worldwide employees, and research and development expenses. Observations that still had missing values in those key variables were ultimately removed from the sample. Additionally, firms that had fewer than five worldwide employees were also dropped. Once the full sample was complete, two subsamples were created based on the number of worldwide employees: a large firm sample (greater than or equal to 500 employees) and a small firm sample (less than 500 employees).[2] To account for innovation differences across industry types, NAICS codes were used to separate each firm into broader business categories. Categories defined in the BRDIS survey were used and are listed below:

[2] All reported total observations and firm counts are rounded approximations based on Census data disclosure policies.

- Aerospace and Defense
- Automobiles, Motorcycles, and Components
- Capital Equipment
- Chemicals and Materials
- Consumer Goods
- Energy and Mining
- Finance, Insurance, and Real Estate
- Healthcare
- Information Technology-Goods and Services
- Professional, Scientific, and Technical Services
- Telecommunications and Utilities
- Other Services

For confidentiality purposes and to meet Census disclosure requirements, some of these industries had to be consolidated. Consolidated categories differed between the full sample and the two subsamples. In the full sample, Capital Equipment and Energy and Mining were consolidated into one category, while Finance, Telecommunications, and Scientific Services were combined. Additionally, a general manufacturing category was created for both the large firm and small firm subsamples, which included the Aerospace, Automotive, Capital Equipment, and Energy and Mining sectors.

4.6. Analytical Strategy

Because of the binary nature of the reported innovation variables, a random effects logistic regression model was used for the analysis. Variables for process innovation and product innovation were coded as zero or one; they are coded as one if product innovation or process

innovation occurred for the given time and zero otherwise. Non-linear logistic regression models are essential for creating predicted values that conform to the binary nature of the variable. Given the unknown differences across the panel sample, a random effects model was used to account for random variations among firms that may impact the likelihood of innovation. Additionally, we opted for a random effects model instead of a fixed effects model for two reasons. First, our sample was drawn from a large population, making it more suitable to treat individual-specific constant terms as randomly distributed across cross sectional units, as suggested by Greene (2000, p. 567). Second, estimates calculated using fixed effects models may exhibit bias when applied to panels covering relatively short periods. This bias is less of a concern with random effects models, as noted by Hsiao (1986) and Heckman (1981). Given that all firm-year observations in our sample spanned just eight years, the random effects model was the more appropriate choice.

The predicted probability is determined by a logistic cumulative distribution function. In a panel with i individuals and t time periods, the probability of a binary outcome Y_{iIt} for the *i*-th individual at time *t* is estimated using the following probability function:

$$P\,(Y_{it} = 1) = exp(X_{it}\beta + u_i) \,/\, (1 + exp(X_{it}\beta + u_i)) \quad (1)$$

where:

exp is the exponential function: $exp(x) = e^x$.

Xit represents a vector of explanatory variables for individual *i* at time *t*.

β is a vector of fixed effect coefficients that are common to all individuals.

u_i is a random effect specific to the *i*-th individual.

The random effect u_i captures individual-specific unobserved heterogeneity that affects the outcome. The distribution of these random effects is $N(0, \sigma^2)$ – normally distributed with mean

zero and variance σ^2. The maximum likelihood estimation (MLE) procedure is used to estimate the model parameters (β and σ^2).

The STATA software program was used for the analysis; specifically, xtlogit was the model command. Equation 2 below shows the equation for the full sample for product innovation. The same model is used for process innovation as well as for the two subsamples based on number of employees. By defining the firm effect as random, it allows us to achieve a greater degree of generalizability.

$$\begin{aligned} &Product\ Innovation\ (0|1) = \alpha + \beta_1 \ln(R\&D\ Expenses) + \\ &\beta_2 \ln(world\ employees) + \beta_3 \ln(patents\ applied) + \beta_4 \ln(patents\ issued) + \\ &\beta_5 R\&D\ New\ Business\ (0|1) + \beta_6 Collaborative\ R\&D\ (0|1) + \\ &\beta_7 R\&D\ from\ Grants\ (0|1) + Industry\ Controls + Year\ Controls \end{aligned} \quad (2)$$

The full sample consisted of approximately 7,500 observations across nine years from roughly 850 unique firms.[3] For the subsample of large firms with 500 or more employees, there were approximately 4,500 observations and about 500 unique firms. For the subsample of small firms with under 500 employees, there were approximately 3,500 observations and about 400 unique firms. Firms could be included in each subsample for different years if the number of employees crossed the threshold at some point.

5. Results

Table 2 presents the results of our statistical analysis. In Models 1 and 2, we examine the relationship between the independent variables and product innovation (Model 1) and process innovation (Model 2) for the entire sample. This analysis aims to demonstrate the effect of firm

[3] All firm counts shown are calculated from the rounded observation counts listed. Census disclosure rules require that all such counts be rounded.

size on innovation outcomes and how the magnitude of the relationships changes when we divide our sample into large firms (500 or more employees) and small firms (less than 500 employees).

In Model 1, we examine the effect of our independent variables on product innovation. When using the entire sample, we find that firm size is not a significant determinant of product innovation. However, patents applied for, R&D toward new business areas, and collaborative R&D are significant determinants of product innovation. In Model 2, firm size emerges as a significant predictor of process innovation, alongside patents applied for, R&D toward new business areas, and collaborative R&D. Thus, we observe a high level of consistency in the determinants of both product innovation and process innovation across the two models, with the only difference pertaining to the effect of firm size. The values of the coefficients for the industry and year control variables had to be suppressed to meet Census review policies.[4]

Table 2. Results from the random effects model

Model	1	2	3	4	5	6
	Full Sample	Full Sample	Large Firms	Large Firms	Small Firms	Small Firms
	Product Innovation	Process Innovation	Product Innovation	Process Innovation	Product Innovation	Process Innovation
Log (R&D Expenses)	0.0622	-0.02918	0.1231	-0.0551	-0.03649	-0.01909
	(-6.98e-02)	(-6.58e-02)	(-9.33e-02)	(-8.14e-02)	(-1.12e-01)	(-1.18e-01)
Log (Worldwide Employees)	0.08226	0.3109	-0.003853	0.4541	0.6319	0.3536
	(-7.36e-02)	(6.763e-02) ***	(-1.19e-01)	(1.043e-01) ***	(1.596e-01) ***	(1.541e-01) **
Automobiles	incl.	incl.				
Aerospace and Defense	incl.	incl.				
Capital Equipment	incl.	incl.				
Chemicals and Materials	incl.	incl.	incl.	incl.	incl.	incl.
Consumer Goods	incl.	incl.	incl.	incl.	incl.	incl.
FIR/Telecom/Sci-services	incl.	incl.	incl.	incl.	incl.	incl.
Healthcare	incl.	incl.	incl.	incl.	incl.	incl.
Information Technology	incl.	incl.	incl.	incl.	incl.	incl.
	0.231	0.105	0.2136	0.05059	0.2554	0.1939

[4] We would like to clarify that we would be able to clear more results and conduct new analyses based on the comments from the review team. The U.S. Census has a specific policy regarding accessing data for revision of the manuscripts submitted to journals for publication that allows us to conduct the necessary analysis.

Model	1	2	3	4	5	6
	Full Sample	Full Sample	Large Firms	Large Firms	Small Firms	Small Firms
	Product Innovation	Process Innovation	Product Innovation	Process Innovation	Product Innovation	Process Innovation
Log (Patents Applied For)	(7.081e-02) ***	(6.223e-02) *	(9.103e-02) **	-7.72e-02	(1.191e-01) **	(1.100e-01) *
Log (Patents Issued)	0.05441	0.07638	0.1613	0.1479	-0.1668	-0.1202
	(-7.65e-02)	(-6.66e-02)	(9.677e-02) *	(8.051e-02) *	(-1.41e-01)	(-1.32e-01)
R&D Toward New Business Areas	0.006243	0.006285	0.00144	0.005646	0.01113	0.007599
	(2.109e-03) ***	(1.902e-03) ***	-2.95e-03	(2.577e-03) **	(3.083e-03) ***	(2.862e-03) ***
Collaborative R&D	0.5102	0.4597	0.8294	0.4921	0.2027	0.4159
	(2.137e-01) **	(1.803e-01) **	(2.917e-01) ***	(2.241e-01) **	(-3.31e-01)	(-3.17e-01)
Grants for R&D	-0.3727	-0.06266	-0.1613	0.1267	-0.6954	-0.6
	(-2.40e-01)	(-2.06e-01)	(-3.13e-01)	(-2.49e-01)	(3.981e-01) *	(-4.01e-01)
2009.year	incl.	incl.	incl.	incl.	incl.	incl.
2010.year	incl.	incl.	incl.	incl.	incl.	incl.
2011.year	incl.	incl.	incl.	incl.	incl.	incl.
2012.year	incl.	incl.	incl.	incl.	incl.	incl.
2013.year	incl.	incl.	incl.	incl.	incl.	incl.
2014.year	incl.	incl.	incl.	incl.	incl.	incl.
2015.year	incl.	incl.	incl.	incl.	incl.	incl.
2016.year	incl.	incl.	incl.	incl.	incl.	incl.
Manufacturing (Aerospace, Auto, Capital Equipment)			incl.	incl.	incl.	incl.
Constant	incl.	incl.	incl.	incl.	incl.	incl.
Observations	7500	7500	4500	4500	3000	3000
Log Likelihood	-2833	-3296	-1595	-1984	-1201	-1287
Wald χ^2	255.9	328.7	190	224.6	114.3	128
Rho	0.77	0.72	0.79	0.73	0.73	0.7
SigmaU	3.29	2.89	3.56	2.97	2.97	2.75

Standard errors in parentheses. * significant at 10%, ** significant at 5%, *** significant at 1%

In Models 3-6, we examine the relationships separately for large firms and small firms. Models 3 and 4 present the results for large firms regarding product innovation and process innovation, respectively. In Model 3, we find that patents applied for, patents issued, and collaborative R&D are significant determinants of product innovation. In Model 4, firm size,

patents issued, R&D toward new business, and collaborative R&D are determinants of process innovation for large firms. Similarly, Models 5 and 6 depict the same relationships for small firms. In Model 5, firm size, patents applied for, R&D toward new business, and grants for R&D are significant determinants of product innovation for small firms. In Model 6, firm size, patents applied for, and R&D toward new business emerge as significant determinants of process innovation for small firms.

Summary of the results. Table 3 presents a high-level summary of the results, which are provided for the entire sample as well as for large firms and small firms. Although there are some differences in the determinants of product innovation and process innovation between the two approaches to data analysis (using the entire sample versus conducting analysis for large firms and small firms separately), firm size has an important impact on the relationship between the innovation activities of the firm and innovation outcomes. While the analysis of the entire sample employs similar determinants for product innovation and process innovation (except for firm size), our analysis using two samples indicates a significant difference between large firms and small firms. Comparing the results for large firms and small firms, patents applied for is the only common determinant of product innovation, while firm size and R&D toward new business are the only common determinants of process innovation. Grants for R&D have a significant impact on product innovation for small firms.

Table 3. Comparative analysis of the results

Variables/Models	Entire Sample		Large Firms		Small Firms	
	Product Innovation	Process Innovation	Product Innovation	Process Innovation	Product Innovation	Process Innovation
R&D						
Firm Size		Yes		Yes	Yes	Yes
Patents applied for	Yes	Yes	Yes		Yes	Yes
Patents issued			Yes	Yes		
R&D toward new business	Yes	Yes		Yes	Yes	Yes

Collaborative R&D	Yes	Yes	Yes	Yes		
Grants for R&D					Yes	

Exploring the relationship between size and innovation. One of the important questions in innovation management is the debate related to the relationship between size and innovation. The literature generally supports the idea that smaller firms emphasize product innovation, and larger firms emphasize process innovation (Prusa and Schmitz, 1991; Pavitt et al., 1987). A comparison of the regression coefficients from the variable for the number of employees for large firms (0.4541) and small firms (0.3536) provides further empirical support. We can further examine the significance of this difference by utilizing a specific statistical test (Clogg et al., 1995; Paternoster et al., 1998):

$$Z = \frac{b_1 - b_2}{\sqrt{SEb_1^2 + SEb_2^2}} \quad (3)$$

In this procedure, b_1 and b_2 are the coefficients of the regression estimates, SEb_1^2 and SEb_2^2 are the variance for the regression estimates for the two groups (small firms and large firms), and Z is a measure of the difference between two regression coefficients. With this procedure, we can determine whether the effect of firm size on process innovation is different between large firms and small firms. The Z statistic follows a normal distribution, so an absolute value of Z above 1.645 is an indication of a significant difference between the regression coefficients for the two samples. Using the data from Table 1, the Z statistic is determined to be 2.90. This suggests that there is a significant difference in the effect of firm size on process innovation between large firms and small firms, and the effect is more pronounced for large firms.

We also examine whether the effect of firm size on product innovation (0.6319) and process innovation (0.3536) is invariant for small firms. Using the procedure discussed above, we determine the Z value for the difference between the coefficients for product innovation and

process innovation. The Z value is 5.65, suggesting that the effect of firm size on product innovation is more significant compared to that of process innovation. Thus, while small firms benefit from both product innovation and process innovation, the effect is more pronounced for product innovation.

6. Discussion

6.1 Theoretical Contributions

The study provides several important contributions to the management of innovation and technology literature. The findings suggest that the determinants of innovation are contingent upon the size of the firm. These findings are aligned with the contingency theory of management, which suggests that the effectiveness of management practices is contingent upon contextual and organizational factors (Geng et al., 2021; Shenkar and Ellis, 2022; Parast, 2022). We provide empirical evidence for the contingency role of firm size on the effectiveness of innovation activities of firms, and we show how specific practices enhance product innovation and process innovation in small versus large firms. In small organizations, firm size is a determinant of both product innovation and process innovation. However, in large organizations, firm size just impacts process innovation. This contingent view to understand the attainment of innovation represents a contribution to literature.

Another contribution of the paper is reflected in the critical role of collaborative R&D in enhancing innovation outcomes. Although studies have discussed the advantages and disadvantages of collaborative R&D on innovation (Bougrain and Haudeville, 2002; Hottenrott and Lopes-Bento, 2016; Bustinza et al., 2019), few studies have examined the conditions under which such relationships hold; in particularly, whether such relationship is contingent upon firm size. We unpack the effect of collaborative R&D on innovation outcomes under different firm

sizes; we show that large firms benefit the most from collaborative R&D. This result builds on the literature that examines R&D collaboration issues in small firms (Narula, 2004; Akinremi and Roper, 2021). The results also build on the literature on innovation management that has hitherto questioned the value that small firms obtain from participating in collaborative R&D projects.

Our finding on the impact of *number of patents applied for* on innovation outcomes builds on prior studies (Meliciani, 2000; Das, 2020). The fact that *patents applied for* is shown to be a significant predictor of product innovation for both small firms and large firms suggests the presence of other activities which help to translate and implement patents into innovation outcomes. Such activities include a myriad of factors that are directly linked to the management and effective utilization of organizational resources (including R&D investment decisions) to ensure that new knowledge developed in form of patents are transformed into innovation outcomes. This finding demonstrates the applicability of the resource orchestration theory in the innovation process; it shows the role of structuring, bundling, and leveraging firm resources in translating patents into innovative outcomes (Baert et al., 2016; Simon et al., 2011; Queiroz et al., 2022).

Our results also contribute empirically to the debate on the effect of firm size on product innovation and process innovation. The general findings reported in the literature suggest that larger firms focus more on process innovation, while smaller firms focus on product innovation (Fritsch and Meschede, 2001). We confirm this empirically in our study; however, we provide additional evidence that smaller firms also focus on process innovation, albeit in comparison the effect is less pronounced than in large firms. The literature alludes to the complementary benefits of both product innovation and process innovation for small and micro-organizations (Vonortas and Xue, 1997; Kamutando and Tregenna, 2023). While there is literature that suggests small

organizations benefit from product innovation, process innovation facilitates the growth strategy in small businesses (Hervas-Oliver et al., 2014).

Pavitt (1985) reported that small firms tend to patent more per unit of R&D than large firms. This has led to the argument put forward by some scholars that small firms tend to exhibit higher levels of innovation compared to large firms, as they achieve more patent counts and citations for every dollar invested in R&D (Plehn-Dujowich, 2009). We provide a different perspective to the relationship between firm size and patenting strategy by differentiating between the effect of patents applied for and patents issued. We found that for small firms, the number of patents applied for is a significant predictor of both product innovation and process innovation, while for large firms, the number of patents issued is the major predictor of product innovation and process innovation. These results provide new insights regarding the value of the patenting system for small businesses. The relationship of patents applied for to both product innovation and process innovation for small firms reflects the potential benefits of applying for patents; more importantly, it suggests that there are mechanisms that facilitate the translation of patents applied for to innovation outcomes. Our study complements a study conducted by Vaona and Pianta (2008) in which they examined the effect of patent applications on product innovation and process innovation between small, medium, and large organizations. They found that the number of patent applications is a significant predictor of product innovation for small and medium-sized organizations but not for large organizations. We provide a more holistic view of the effect of a firm's patenting strategy on innovation outcomes by making a distinction between the effects of patents applied for and patents issued and how this differs by firm size, thus contributing to the extant innovation literature.

Our study underlines the critical role of *collaborative R&D* for large firms in enhancing both product innovation and process innovation. The literature provides mixed findings regarding the effectiveness of collaborative R&D. While some studies suggest that such partnerships are useful in technology acquisition and technology development (Huikkola et al., 2013; Melnychuk et al., 2021), other studies report that such endeavors fail because of difficulties in managing the partnerships and the complexity of such relationships (Lhuillery and Pfister, 2009; Hottenrott and Lopes-Bento, 2016; Guzzini et al., 2018). While many factors contribute to the success or failure of any form of partnership, our results provide new insight into how benefits from collaborative R&D differ by firm size. Specifically, such partnerships provide better outcomes for larger organizations, while smaller organizations do not gain much benefit from these relationships. One possible explanation for the effectiveness of collaborative R&D for large organizations is the greater availability of resources, which allows them to allocate sufficient resources to exploit such partnerships, minimize risk and mitigate potential problems during the collaborative R&D efforts.

Our empirical results show that R&D investment, by itself, is not a significant determinant of innovation outcomes. Indeed, we did not find a significant relationship of R&D investment to product innovation or process innovation in either small or large organizations. Rather, we found that the outcomes of R&D investment such as patents applied for, patents issued, and collaborative R&D are significant determinants of innovation outcomes. This result contributes to the literature and is in line with the academic and practitioners' literature that there is clear distinction between having a successful R&D process and achieving market success with products, and real innovation needs much more that an R&D budget (Wolff, 2007; Wang et al., 2010; Viki (2016; O'Connor, 2019).

6.2. Managerial Implications

This study provides important insights for R&D managers, innovation and technology managers, and operations managers. The findings suggest similarities and differences between small firms and large firms in their innovation management practices and outcomes. For instance, small firms may benefit more from agile innovation processes and collaborative partnerships, whereas large firms may focus on scaling up successful innovations and investing in advanced R&D capabilities. Therefore, organizations should tailor their innovation practices based on firm size to enhance both product and process innovation.

The dynamics of innovation in small firms and large firms have important implications for managing supply chains. Since a firm may work with suppliers of varying size (small and large), the innovation capability of the firm is tightly intertwined with the innovation capability of its suppliers and how a supplier can respond to a new product, a new service, or process of innovation of the firm. This requires a focal firm to understand the nuances related to the dynamics of innovation in small firms and large firms, to better align its expectations of suppliers with their capacities to achieve desired innovation outcomes. A small firm that is on a path to growth might be able to address both process innovation and product innovation more effectively than a small firm that is not pursuing such a strategy. This has important implications for buyer-supplier relationships and how firms, small or large, evaluate their supplier selection decisions and manage their relationships (Pearson and Ellram, 1995; Koufteros et al., 2007; Adams et al., 2012). So, one important insight for supply chain managers would be the selection of suppliers based on their innovation capabilities, and the extent to which the firm needs complementary skills in product innovation, process innovation, or both from their suppliers. Supply chain managers should be cognizant of such contextual factors when working with their suppliers when supply chain innovation efforts are concerned with product innovation or process innovation.

The significant role of collaborative R&D for large organizations underscores the importance of effective management practices at all levels within the organizations. Specifically, management of collaboration activities necessitates attention to practices such as fostering inter-firm trust, integrating information, and coordinating processes, as these have been shown to enhance a firm's competitive advantage (Blomqvist et al., 2005; Bunduchi, 2013). For R&D and innovation managers, it is crucial to incorporate best collaborative practices into their R&D programs. This includes implementing strategies such as project management, scheduling, quality control, resource planning, and allocation, which are rooted in operations management principles. By integrating these practices into R&D management efforts, managers can ensure the successful outcomes of collaborative R&D projects.

Our study demonstrates that firms benefit from their patenting strategy, contingent upon their size. This finding has significant implications for policymakers, suggesting the need to tailor interventions to specific firm groups. Efforts aimed at stimulating innovation should identify the major groups, distinguishing between large and small firms, and devise strategies to encourage patenting within these categories. Policymakers may consider implementing measures to streamline the patenting process, making it more accessible and less burdensome for firms of all sizes. Additionally, initiatives focused on training and education, emphasizing the legal aspects of patenting, could be implemented to equip firms with the knowledge and skills necessary to navigate the patenting landscape effectively.

6.3. Limitations and Future Research

Similar to many studies, our research is subject to several limitations. One significant limitation is related to our measure of product innovation and process innovation, which were dichotomous (indicating whether a firm successfully introduced a product/service innovation or a

process innovation). Unfortunately, the Business R&D and Innovation Survey (BRDIS) does not provide information on the number of new products/services or new processes developed by organizations; it only indicates whether firms developed new products/services or processes. This limitation restricts the predictive capability of our models, making it more challenging to accurately determine the strength of the relationships between the independent and dependent variables. It would have been helpful if the numbers of product innovations and process innovations were reported.[5]

Another limitation of the study pertains to the limited information available regarding the management practices within organizations. While R&D investment plays a crucial role in the development of new products and processes, it is ultimately the management systems that shape the innovation capabilities of an organization. Unfortunately, the Business R&D and Innovation Survey (BRDIS) provides limited insight into organizational management practices. Future studies could enhance their analyses by incorporating firm-level management practices to conduct a more nuanced assessment of innovation practices within firms. Furthermore, given that a firm's innovation performance is influenced by its supply chain, integrating variables at the supply chain level could bolster the validity and robustness of the findings. We suggest that future research consider incorporating operational measures of firm performance alongside supply chain characteristics to provide a more comprehensive understanding of the relationship between innovation and performance.

Furthermore, it will be crucial to investigate the relationship between innovation outcomes and a firm's financial performance. While our study's innovation measures capture the actual

[5] As part of the requirements of using the federally protected data, the researchers are required to inform the U.S. Census about shortcomings of the data The researchers involved in this study informed the U.S. Census Bureau on this shortcoming.

innovation within organizations (i.e., the commercialization of products and processes), it can be assumed that there has been a benefit-cost assessment regarding their effectiveness. However, linking innovation outcomes to financial performance offers a more comprehensive understanding of how resources are transformed into new knowledge that can generate the highest financial returns. Given the divergent innovation strategies between small and large firms, such findings would provide valuable insights for R&D managers, enabling them to prioritize their innovation investment decisions more effectively.

Lastly, an important direction for future studies is to investigate the effectiveness of R&D and innovation management practices specifically tailored for micro-organizations (Inan and Bititci, 2015; Faherty and Stephens, 2018; Yáñez-Araque et al., 2021), which are defined as organizations with fewer than 10 employees. The dynamics within micro-organizations appear to differ from those of small organizations examined in this study. Considering that management practices often vary based on firm size, it would be beneficial to explore how innovation management practices are formulated within micro-organizations and how they distinguish themselves from those adopted by small and large organizations.

References

Adams, J. H., Khoja, F. M., and Kauffman, R. (2012) An empirical study of buyer–supplier relationships within small business organizations, Journal of Small Business Management. 50(1), 20-40.

Athreye, S. S., Fassio, C., and Roper, S. (2021) Small firms and patenting revisited, Small Business Economics, 57, 513–530.

Akinremi, T. and Roper, S. (2021). The Collaboration Paradox: Why Small Firms Fail to Collaborate for Innovation. In: Fernandes, G., Dooley, L., O'Sullivan, D., Rolstadås, A. (eds) Managing Collaborative R&D Projects. Contributions to Management Science. Springer, Cham.

Alvarez, R., and Crespi, G. (2003) Determinants of technical efficiency in small firms. Small Business Economics, 20, 233–244.

Andries, P., and Faems, D. (2013) Patenting activities and firm performance: Does firm size matter? Journal of Product Innovation Management, 30(6), 1089-1098.

Andries, P. and Faems, D. (2013) Patenting activities and firm performance: does firm size matter? 30(6), 089-1098.

Artz, K. W., Norman, P. M., Hatfield, D. E., and Cardinal, L. B. (2010) A longitudinal study of the impact of R&D, patents, and product innovations on firm performance, Journal of Product Innovation Management, 27(5), 725-740.

Aw, B.-Y. and Batra, G. (1998) Firm size and the pattern of diversification, International Journal of Industrial Organization, 16(3), 313-331.

Baert, C., Meuleman, M., Deepuyne, M., and Wright, M. (2016) Portfolio entrepreneurship and resource orchestration, Strategic Entrepreneurship Journal, 10, 346–370.

Barney, J. (1991). Firm resources and sustained competitive advantage. Journal of Management, 17(1), 99-120.

Bauer, M. and Leker, J. (2013) Exploration and exploitation in product and process innovation in the chemical industry, R&D Management, 43 (3), 196-212.

Baumann, J. and Kritikos. A. S. (2016) The link between R&D, innovation and productivity: are micro firms different? Research Policy, 45 (6), 1263–1274.

Blomqvist, K., Hurmelinna, P., and Seppänen, R. (2005) Playing the collaboration game right—balancing trust and contracting, Technovation, 25(5), 497-504.

Berends, H., Jelinek, M., Reymen, I., and Stultiëns, R. (2014) Product innovation processes in small firms: combining entrepreneurial effectuation and managerial causation, Journal of Product Innovation Management, 31 (3), 616-663.

Bougrain, F. and Haudeville, B. (2002) Innovation, collaboration and SMEs internal research capacities. Research Policy, 31, 735-747.

Braguinsky, S., Ohyama, A., Okazaki, T., and Syverson, C. (2021) Product innovation, product diversification, and firm growth: evidence from Japan's early industrialization, American Economic Review, 111(12), 3795-3826.

Burns, T. and Stalker, G. M. (1961) The Management of Innovation. Chicago, IL: Quadrangle Books.

Bunduchi, R. (2013) Trust, partner selection and innovation outcome in collaborative new product development, Production Planning & Control, 24(2-3), 145-157.

Bustinza, O. F., Gomes, E., Vendrell-Herrero, F., and Baines, T. (2019) Product–service innovation and performance: the role of collaborative partnerships and R&D intensity, R&D Management, 49(1), 33-45.

Camison-Zornoza, C., Lapiedra-Alcami, R., Segarra-Cipres, M., and Boronat-Navarro, M. (2004) A meta-analysis of innovation and organizational size, Organization Studies, 25(3), 331–361.

Carnes, C. M., Hitt, M. A., Sirmon, D. G., Chirico, F., and Huh, D. W. (2022) Leveraging resources for innovation: The role of synchronization, Journal of Product Innovation Management, 39 (2), 160-176.

Cesaroni, F. and Piccaluga, A. (2013) Operational challenges and ST's proposed solutions to improve collaboration between IP and R&D in innovation processes, California Management Review, (4), 143-156.

Chandler, A. D. and Chandler, A. D. Jr (1962) Strategy and structure: Chapters in the history of the industrial enterprise, The American Historical Review, 68, 158–60.

Cheng, C.C.J. and Sheu, C. (2024) Social media analytics and product innovation: mediating effects of knowledge exploration and exploitation competences, International Journal of Operations & Production Management, 44(1), 229-259.

Chirumalla, K. (2021) Building digitally-enabled process innovation in the process industries: A dynamic capabilities approach, Technovation, 105, 102256.

Clogg, C. C., Petkova, E. and Haritou A. (1995) Statistical methods for comparing regression coefficients between models. American Journal of Sociology, 100, 1261-1293.

Choo, A., Narayanan, S., Srinivasan, R., and Sarkar, S. (2021) Introducing goods innovation, service innovation, or both? Investigating the tension in managing innovation revenue streams for manufacturing and service firms, Journal of Operations Management, 67(6), 704-728.

Cohen, W. M., and Klepper, S. (1996a) A reprise of size and R&D, Economic Journal, 106, 925-951.

Cohen, W. M., and Klepper, S. (1996b) Firm Size and the nature of Innovation within Industries: The Case of Process and Product R&D, The Review of Economics and Statistics, 78 (2), 232-243.

Crossan, M.M. and Apaydin, M. (2010) A multi-dimensional framework of organizational innovation: A systematic review, Journal of Management Studies, 47 (6), pp. 1154-91.

Cozzarin, B. P. (2017) Impact of organizational innovation on product and process innovation, Economics of Innovation & New Technology, 26 (5), 405-417.

Damanpour, F. (1991) Organizational innovation–a meta-analysis of effects of determinants and moderators, Academy of Management Journal, 34 (3), 555–590.

Damanpour, F. (1992) Organizational size and innovation, Organization Studies, 13 (3), 375–402.

Damanpour, F. and Gopalakrishnan, S. (2001) The dynamics of the adoption of product and process innovations in organizations, Journal of Management Studies, 38 (1), 45-65.

Damanpour, F. and Aravind, D. (2006) Product and process innovations: A review of organizational and environmental determinants, in J. Hage and Meeus, M (eds), Innovation, Science, and Institutional Change: A Research Handbook, pp. 38-66, Oxford University Press, Oxford, UK.

Das, R.C. (2020) Interplays among R&D spending, patent and income growth: new empirical evidence from the panel of countries and groups. Journal of Innovation and Entrepreneurship 9(18), 1-22.

De Jong, P. J. J. and. Marsili, O. (2006) The fruit flies of innovations: A taxonomy of innovative small firms, Research Policy, 35, 213-229.

Ding, B., Hernández, X. F., and Jané, N. A. (2023) Combining lean and agile manufacturing competitive advantages through Industry 4.0 technologies: an integrative approach, Production Planning & Control, 34(5), 442-458.

Downs, G. W. and L. B. Mohr, L. B. (1976) Conceptual issues in the study of innovation, Administrative Science Quarterly, 21 (4), 700–714.

Faherty U. and Stephens, S. (2016) Innovation in micro enterprises: reality or fiction?, Journal of Small Business and Enterprise Development, 23 (2), 349 – 362.

Frank, A.G., Thürer, M., Godinho Filho, M. and Marodin, G.A. (2024) Beyond Industry 4.0 – integrating Lean, digital technologies and people, International Journal of Operations & Production Management (in press)

Freel, M. (2003) Sectoral patterns of small firm innovation, networking and proximity, Research Policy, 32, 751–770.

Gao, J., Zhang, W., Guan, T., Feng, Q., and Mardani, A. (2023) The effect of manufacturing agent heterogeneity on enterprise innovation performance and competitive advantage in the era of digital transformation, Journal of Business Research, 155 (Part A), 113387.

Geng, D., Lai, K.-h, and Zhu, Q. (2021) Eco-innovation and its role for performance improvement among Chinese small and medium-sized manufacturing enterprises, International Journal of Production Economics, 231, 107869.

Greene, W. (2000) Econometric Analysis, 4th ed. Prentice Hall, Upper Saddle River, NJ.

Fritsch, M. and Meschede, M. (2001) Product innovation, process innovation, and size. Review of Industrial Organization 19, 335–350.

Ghosh, S. (2009) R&D in Indian Manufacturing Enterprises: What Shapes it? Economics of Innovation and New Technology, 18 (4):337–352.

Griffin. A. (1997) PDMA research on new product development practices: Updating trends and benchmarking best practices, Journal of Product Innovation Management, 14 (6), 429-458.

Guzzini, E., Iacobucci D., and Palestrini A. (2018) Collaboration for innovation and project failure. A dynamic analysis, Economics of Innovation and New Technology, 27(8), 695-708.

Heckman, J. (1981) Statistical Models for Discrete Panel Data," in The Econometrics of Panel Data. McFadden, D. and Manski, C. (Eds), MIT Press, Cambridge MA, 114–178.

Hervas-Oliver, J. L., Sempere-Ripoll, F., and Boronat-Moll, C. (2014) Process innovation strategy in SMEs, organizational innovation and performance: a misleading debate? Small Business Economics, 43, 873–886.

Hoang, H. and Rothaermel, F. T. (2010) Leveraging internal and external experience: exploration, exploitation, and R&D project performance, Strategic Management Journal, 31(7), 734-758.

Hogan, T. and Chirichiello, J. (1974) The role of R&D in small firms, Research Management, 17(6), 26-30.

Hottenrott, H. and Lopes-Bento, C. (2016) R&D partnerships and innovation performance: Can there be too much of a good thing? Journal of Product Innovation Management, 33(6), 773-794.

Hsiao, C. (1986). Analysis of Panel Data., Cambridge University Press, New York, NY.

Huikkola, T., Ylimäki, J., and Kohtamäki, M. (2013) Joint learning in R&D collaborations and the facilitating relational practices, Industrial Marketing Management, 42(7), 1167-1180.

Inan, G. G. and Bititci, U. S. (2015) Understanding organizational capabilities and dynamic capabilities in the context of micro enterprises: A research agenda, Procedia - Social and Behavioral Sciences, 210, 310-319,

Josefy, M., Kuban, S. Ireland, R.D., and Michael A. Hitt, M. A. (2016) All things great and small: organizational size, boundaries of the firm, and a changing environment, Academy of Management Annals, 9(1), 715-802.

Jajja, M. S. S., Kannan, V. R., Brah, S. A., and Hassan, S. Z. (2017) Linkages between firm innovation strategy, suppliers, product innovation, and business performance, International Journal of Operations & Production Management, 37 (8), 1054-1075.

Kamutando, G. and Tregenna, F. (2023) Complementarity between product and process innovation in small and micro-enterprises in Johannesburg, South Africa, Economics of Innovation and New Technology (in press)

Kesidou, E., Narasimhan, R., Ozusaglam, S. and Wong, C.Y. (2022) Dynamic openness for network-enabled product and process innovation: a panel-data analysis, International Journal of Operations & Production Management, 42(3), 257-279.

Kim, D.-Y, Kumar, V., and Kumar, U. (2012) Relationship between quality management practices and innovation, Journal of Operations Management, 30 (4), 295-315.

Kimberly, J. R. and Evanisko, M. (1981) Organizational innovation: The influence of individual, organizational, and contextual factors on hospital adoption of technological and administrative innovations, Academy of Management Journal, 24 (4), 679–713.

Koufteros, X. A., Edwin Cheng, T. C., and Lai, K. -H. (2007) "Black-box" and "gray-box" supplier integration in product development: Antecedents, consequences and the moderating role of firm size, Journal of Operations Management, 25 (4), 847-870.

Knott, A. M. and Vieregger, C. (2020) Reconciling the firm size and innovation puzzle, Organization Science, 31(2), 477-488.

Leal-Rodríguez, A. L., Eldridge, S., Roldán, J. L., Leal-Millán, A. G., and Ortega-Gutiérrez, J. (2015) Organizational unlearning, innovation outcomes, and performance: The moderating effect of firm size, Journal of Business Research, 68(4), 803-809.

Lhuillery, S. and Pfister, E. (2009) R&D cooperation and failures in innovation projects: Empirical evidence from French CIS data, Research Policy, 38 (1), 45-57.

Lee, G. and Xia, W. (2006) Organizational size and IT innovation adoption: A meta-analysis, Information & Management, 43(8), 975-985.

Licht, G. and Zoz, K. (2000) Patents and R&D An Econometric Investigation Using Applications for German, European and US Patents by German Companies, In: The Economics and Econometrics of Innovation. Springer, Boston, MA. https://doi.org/10.1007/978-1-4757-3194-1_12

LiPuma, J. A., Newbert, S. L., and Doh, J. P. (2013) The effect of institutional quality on firm export performance in emerging economies: a contingency model of firm age and size, Small Business Economics, 40, 817–841.

Lundvall, K. and Battese, G. E. (2000) Firm size, age and efficiency: Evidence from Kenyan manufacturing firms, The Journal of Development Studies, 36:3, 146-163.

Marcelino-Sádaba, S., Pérez-Ezcurdia, A., cheverría Lazcano, A. M., and Villanueva, P. (2014) Project risk management methodology for small firms, International Journal of Project Management, 32(2), 327-340.

Mariani, M. M., Machado, I., and Nambisan, S. (2023) Types of innovation and artificial intelligence: A systematic quantitative literature review and research agenda, Journal of Business Research, 155, Part B. 113364.

Mabenge, B. K., Ngorora-Madzimure, G. P. K., and Charles Makanyeza, C. (2022) Dimensions of innovation and their effects on the performance of small and medium enterprises: the moderating role of firm's age and size, Journal of Small Business & Entrepreneurship, 34(6), 684-70.

Melnychuk, T., Schultz, C., and Wirsich, A. (2021) The effects of university–industry collaboration in preclinical research on pharmaceutical firms' R&D performance: Absorptive capacity's role, Journal of Product Innovation Management, 38 (3), 355-378.

Meliciani, V. (2000) The relationship between R&D, investment and patents: a panel data analysis. Applied Economics, 32(11), 1429–1437.

Mc Namara, L. and Baden-Fuller, C. (2007) Shareholder returns and the exploration–exploitation dilemma: R&D announcements by biotechnology firms, Research Policy, 36(4), 548-565.

Narula, R. (2004) R&D collaboration by SMEs: new opportunities and limitations in the face of globalisation, Technovation, 24 (2), 153-161.

O'Connor, G. (2019) Real Innovation Requires More Than an R&D Budget, Harvard Business Review, December, 1–4.

Orlando, B., Renzi, A., Sancetta, G., and Cucari, N. (2018) How does firm diversification impact innovation? Technology Analysis & Strategic Management, 30(4), 391-404.

Parast, M. M. (2022) Toward a contingency perspective of organizational and supply chain resilience, International Journal of Production Economics, 250, 108667.

Parast, M. M. (2020) The impact of R&D investment on mitigating supply chain disruptions: Empirical evidence from U.S. firms, International Journal of Production Economics, 227, 107671.

Parast, M. M. (2011) The effect of Six Sigma projects on innovation and firm performance, International Journal of Project Management, 29(1), 45-55.

Pavitt, K, Robso, M, and Townsend, J (1987) The size distribution of innovating firms in the UK: 1945-1983, Journal of Industrial Economics, 35(3):297–316.

Pavitt, K. (1985), Patent statistics as indicators of innovative activities: possibilities and problems, Scientometrics, 7, 77-99.

Pearce, C. L. and Ensley, M. D. (2004) A reciprocal and longitudinal investigation of the innovation process: The central role of shared vision in product and process innovation teams (PPITs), Journal of Organizational Behavior, 25 (2), 259-278.

Pearson, J. N. and Ellram, L. M. (1995) Supplier selection and evaluation in small versus large electronics firms, Journal of Small Business Management, 33(4), 53-61.

Prusa, T. J, and Schmitz, J. A., Jr. (1991) Are new firms an important source of innovation? Economic Letters, 35(3):339–342.

Paternoster, R., Brame, R., Mazerolle, P. and Piquero, A. (1998) Using the current statistical test for the equality of regression coefficients, Criminology, 36 (4), 859-866.

Plehn-Dujowich, J M. (2009) Firm size and types of innovation, Economics of Innovation and New Technology, 18:3, 205-223.

Prajogo, D.I. and Oke, A. (2016) Human capital, service innovation advantage, and business performance: The moderating roles of dynamic and competitive environments, International Journal of Operations & Production Management, 36(9), 974-994.

Queiroz, M. M., Wamba, S. F., Chiappetta Jabbour, C. J., and Machado, M. C. (2022) Supply chain resilience in the UK during the coronavirus pandemic: A resource orchestration perspective, International Journal of Production Economics, 245, 108405.

Revilla, A. J. and Fernández, Z (2012) The relation between firm size and R&D productivity in different technological regimes, Technovation, 32 (11), 609-623.

Rosenbusch, N., Brinckmann, J., and Bausch, A (2011) Is innovation always beneficial? A meta-analysis of the relationship between innovation and performance in SMEs, Journal of Business Venturing, 26 (4), pp. 441-457.

Sabahi, S. and Parast, M. (2020) Firm innovation and supply chain resilience: a dynamic capability perspective, International Journal of Logistics Research and Application, 23(3), 254-269.

Sasidharan, S. and V. Kathuria. (2011) Foreign direct investment and R&D: substitutes or complements – a case Study of Indian manufacturing after 1991 reforms. World Development, 39 (7), 1226–1239.

Shaver, J. M. and Mezias, J. M. (2009) Diseconomies of managing in acquisitions: evidence from civil lawsuits, organization Science, 20 (1), 206-222.

Shefer, D. and Frenkel, A. (2005) R&D, firm size and innovation: an empirical analysis, Technovation, 25 (1), 25-32.

Shenkar, O. and Ellis, S. (2022) The rise and fall of structural contingency theory: A theory's 'autopsy', Journal of Management Studies, 59(3), 782-818.

Schumpeter, J. A. (1934) The Theory of Economic Development, Harvard University Press, Cambridge, MA.

Schumpeter, J. A. (1950) Capitalism, Socialism, and Democracy, Harper and Row, New York, NY.

Singh, D., Khamba, J. S., and Nanda, T. (2017) Influence of technological innovation on performance of small manufacturing companies, International Journal of Productivity & Performance Management, 66 (7), 838-856.

Sirmon, D., Hitt, M., and Ireland, R. D. (2007). Managing firm resources in dynamic environments to create value: Looking inside the black box. Academy of Management Review, 32: 273-292.

Sirmon, D. G., Hitt, M. A., Ireland, R. D., and Gilbert, B. A. (2011) Resource orchestration to create competitive advantage: breadth, depth, and life cycle effects, Journal of Management, 37(5), 1390-1412.

Soderbom, M. and Teal, F. (2004) Size and efficiency in African manufacturing firms: Evidence from firm-level panel data. Journal of Development Economics, 73, 369–394.

Storey, J., Quintas, P., Taylor, P., and Fowle, W. (2002) Flexible employment contracts and their implications for product and process innovation, International Journal of Human Resource Management, 13 (1), 1–18.

Tushman, M. and Nadler, (1986) Organizing for innovation, California Management Review, 28 (3), 74-92.

Vaona, A. and Pianta, M. (2008) Firm size and innovation in European manufacturing, Small Business Economics, 30, 283–299.

Viki, T. (2016) Why R&D Spending Is Not A Measure Of Innovation, Forbes, August 21. https://www.forbes.com/sites/tendayiviki/2016/08/21/why-rd-spending-is-not-a-measure-of-innovation/?sh=1dc9464c77dd

Visnjic, I., Wiengarten, F., and Neely, A. (2016) Only the brave: product innovation, service business model innovation, and their impact on performance, Journal of Product Innovation Management, 33 (1), 36-52.

Vlaisavljevic, V., Gopalakrishnan, S., Zhang, H., Cabello-Medina, C., Guilbault, M. (2022) Dancing with wolves: how R&D human capital can benefit from coopetition, R&D Management, 52(3), 449-464.

Voss, C. S. (1994) Significant issues for the future of product innovation, Journal of Product Innovation Management, 11 (5), 460–463.

Vonortas, N. S. and Xue, L. (1997) Process innovation in small firms: Case studies on CNC machine tools, Technovation, 17 (8), 427-438.

Wang, S., Guidice, R. M., Tansky, J. W., and Wang, Z.-M. (2010) When R&D spending is not enough: The critical role of culture when you really want to innovate, Human Resource Management, 49 (4), 767-792.

Wagner, E.R. and Hansen, E.N. (2005) Innovation in large versus small companies: insights from the US wood products industry, Management Decision, 4(6), 837-850.

Wang, C., Qureshi, I., Feng Guo, F., Qingpu Zhang, Q. (2022) Corporate social responsibility and disruptive innovation: The moderating effects of environmental turbulence, Journal of Business Research, 139, 435-1450.

Wells, R. M. (2008) The product innovation process: Are managing information flows and cross-functional collaboration key? Academy of Management Perspectives, 22 (1), 58-60.

Wickert, C., Scherer, A. G., and Spence, L. J. (2016) Walking and talking corporate social responsibility: implications of firm size and organizational cost, Journal of Management Studies, 53(7), 1169-1196.

Wolff, M. F. (2007) Forget R&D spending-think innovation, Research Technology Management, 50(2), 7-9.

Woodward, J. (1958). Management and Technology. London: HM Stationary Office.

Xie, X., Huo, J., and Zou, H. (2019) Green process innovation, green product innovation, and corporate financial performance: A content analysis method, Journal of Business Research, 101, 697-706.

Yáñez-Araque, B., Sánchez-Infante Hernández, J. P., Gutiérrez-Broncano, S., and Jiménez-Estévez, P. (2021) Corporate social responsibility in micro-, small- and medium-sized enterprises: Multigroup analysis of family vs. nonfamily firms, Journal of Business Research, 124, 581-592.

Zhong, X., Chen, W., and Ren, G. (2022) The effects of performance shortfalls on firms' exploitation and exploration R&D internationalization decisions: Does industry environmental matter?, Technovation, 112, 102408.